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INTRODUCTION

Personal financial management has become a necessity in today's digital-first world, where individuals handle multiple expenses, investments, liabilities, and budgets. However, many people struggle to maintain accurate financial records, analyse spending behaviour, and plan for future liabilities or investments due to the absence of an easy-to-use, insightful financial management tool.

This project, titled “**FinTrack — Mini Finance App**”, aims to help users seamlessly track their income, expenses, investments in mutual funds, and liabilities in a single intuitive interface. Designed with a clean UI/UX, the system allows users to monitor their financial health, get automatic projections of their mutual fund portfolio, and evaluate whether current savings and mutual fund value can cover upcoming liabilities—based on due dates, priorities, and MF allowances.

The project is built using HTML, CSS, JavaScript (vanilla), and minimal React (for theme toggle and visual charts). It employs a modular JavaScript architecture, separating storage, models, UI rendering, and logic into structured JS files.

FinTrack stands out due to its smart calculations, including:

- Coverage analysis for liabilities using current balance, planned incomes, and mutual fund future value.
- SIP projection based on existing mutual fund portfolio units, categories, and expected annual returns, without requiring historical NAV data.
- Auto-updating KPIs, pie charts for portfolio allocation, and category-wise transactions.

This project helps individuals achieve better financial awareness, planning discipline, and decision-making using visual and analytical tools without any external finance platforms.

PROBLEM STATEMENT

In today's digital era, managing personal finances has become increasingly complex due to multiple income sources, rising expenses, investment options, and financial obligations. Individuals, especially students and young professionals, struggle to track their day-to-day transactions, monitor investment performance, plan for future liabilities, and analyse savings effectively. The lack of awareness, poor financial discipline, and unorganized record-keeping often lead to financial instability, overspending, unfulfilled goals, and poor decision-making.

Many existing financial applications are either too complicated, overly professional, or mainly focused on specific areas like investments or budgeting. They often fail to provide a **holistic** view of an individual's finances into a single interface. Additionally, most platforms require technical financial knowledge and are cluttered with complex dashboards, making them unsuitable for beginners or non-finance users.

Users today expect a clean, intuitive, and interactive financial dashboard that allows them to record, visualize, and analyse their finances in real-time. However, most existing tools do not offer features like automatic balance calculations, expense categorization, goal tracking, liability management, SIP projection, or integration of investments with liabilities. As a result, users are unable to clearly understand their financial health, savings capacity, and future planning.

Another major challenge is the lack of automation and decision support. People often find it difficult to analyse whether they can pay off liabilities from savings, allocate money for goals, or predict how current mutual fund investments may grow with systematic investments (SIP). Without proper analysis, users face difficulty in planning budgets, scheduling payments, and achieving financial goals in time.

This project, FinTrack, aims to bridge this gap by developing a simple, user-friendly, and data-driven Personal Finance Tracking System using HTML, CSS, JavaScript, and React.js (for theme and charts). It provides features like transaction management, investment tracking, liability monitoring, SIP projection, and dynamic data visualization using charts.

FinTrack focuses on usability, affordability, transparency, and real-time financial clarity, allowing users to manage and monitor their finances without any technical or accounting knowledge. By bringing together all the financial aspects – Transactions, Liabilities, Investments - into a single interactive dashboard, FinTrack empowers users to take control of their financial decisions confidently.

SPECIFIC REQUIREMENTS

a) Functional Requirements

i. Income and Expense Management:

- Add, filter, sort, search transactions.
- Auto-calculate balance, income, and expense totals.
- Export monthly or yearly summaries to CSV.

ii. Investment Portfolio Management:

- Add mutual fund purchases with scheme, category, units (auto calculated), NAV, and date.
- Update NAV manually to compute current value.
- Pie-chart visualization of category-wise allocation.
- Display total invested value, current value, and wealth change.

iii. SIP Projection & Portfolio Growth Estimation:

- Instead of taking investment amount as input, it uses existing portfolio units, expected category-wise annual returns, and time duration to calculate projected future value.
- Shows fund-wise future value, total growth, and estimated gain.

iv. Liability Coverage Analysis:

- Add liabilities with amount, due date, priority, and whether MF can be used.
- Calculate whether liabilities are covered or short, using:
 - Current balance
 - Planned incomes (by date)
 - Mutual fund value at due date (based on units bought before that date)

v. Smart UI Rendering:

- Reflects data dynamically across tabs.
- Auto-updates balance, liabilities coverage, investments, and KPIs.

b) Non-Functional Requirements:

i) Scalability:

Although FinTrack is currently intended for personal users and runs entirely in the browser using Local Storage, it is designed with modular data structures that can be easily extended to support multiple users, cloud-based storage (Firebase/API).

ii) Performance:

The system ensures fast and responsive interactions, even when handling large datasets of transactions or investment records. Operations like filtering, sorting, searching, and SIP calculations execute efficiently with optimized JavaScript algorithms. Page load times remain minimal due to lightweight frontend architecture (HTML, CSS, Vanilla JS) and absence of heavy backend processing.

iii) Reliability and Availability:

FinTrack operates entirely on the client side using Local Storage, eliminating downtime and external dependency. The system remains available even without Internet connectivity. All previously entered financial data (transactions, liabilities, investments) is automatically preserved across browser sessions. Error handling ensures that incorrect entries do not affect system stability.

iv) Usability and Accessibility:

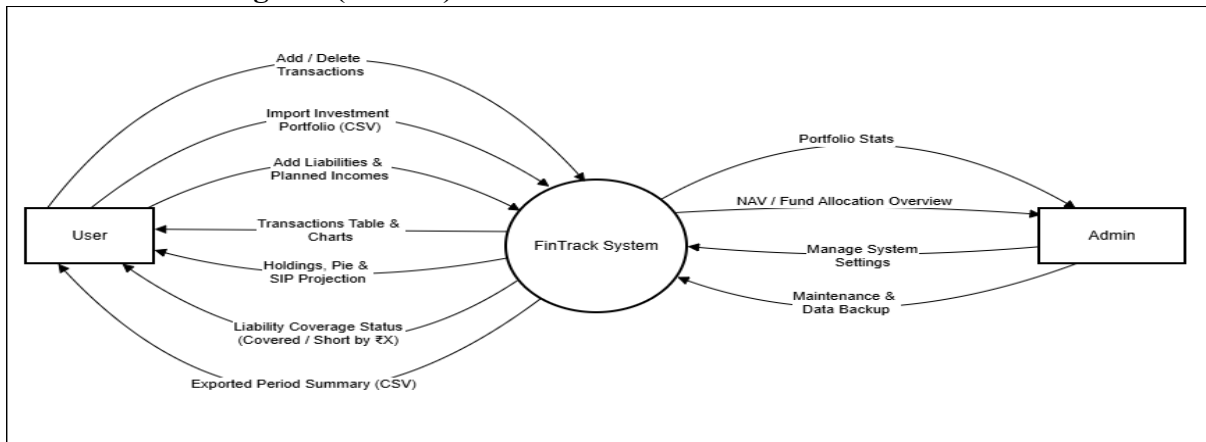
The interface is kept simple, clean, and user-friendly. Tabs such as Transactions, Investments, and Liabilities provide clear and intuitive navigation. Buttons, labels, placeholders, and form fields are easily understandable, even for non-technical users. The application can be used comfortably by students, self-employed individuals, or casual users with no finance background.

v) Offline Support:

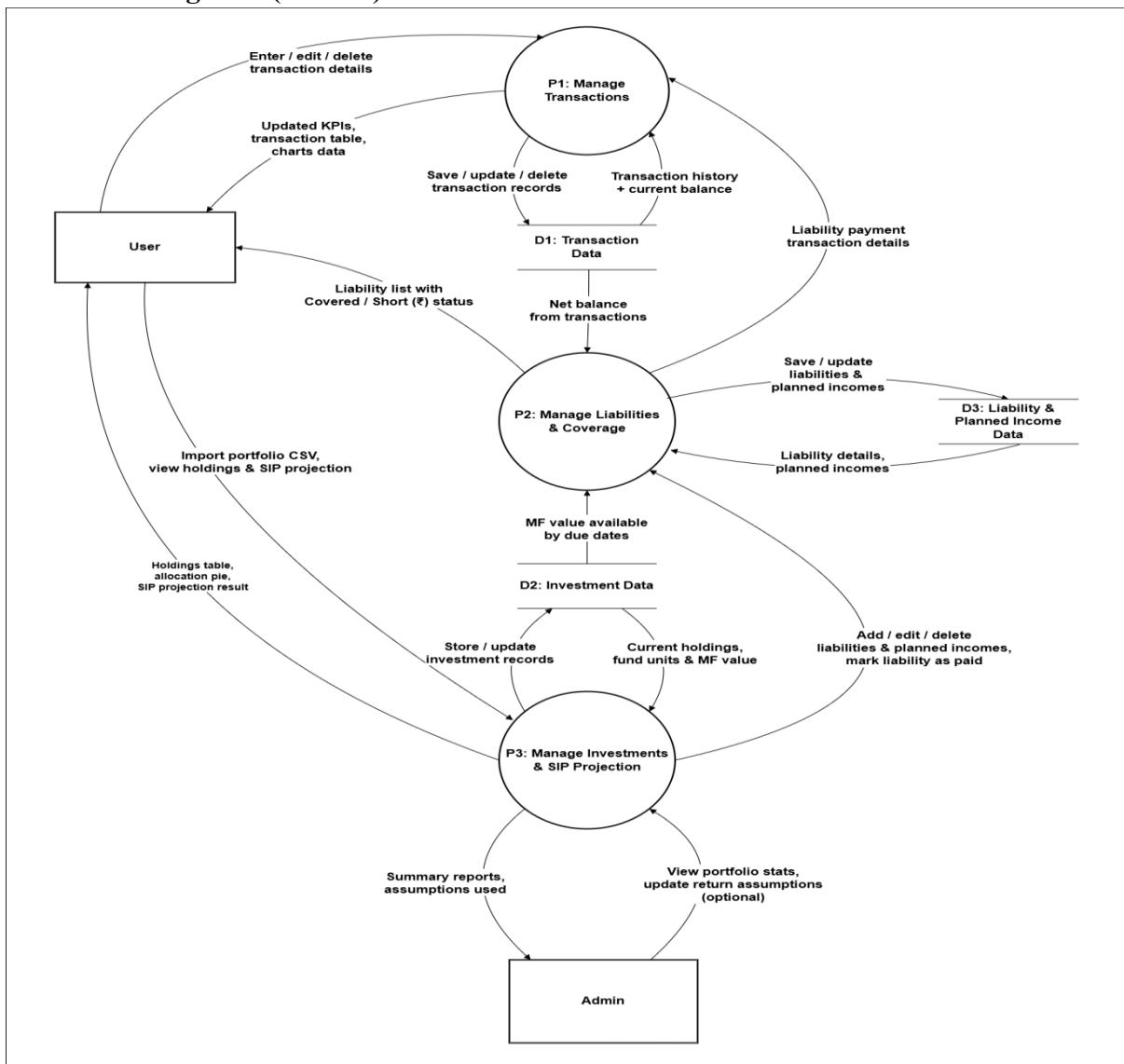
All features, including adding transactions, viewing balance, calculating future SIP, or liability coverage, work fully offline. No Internet connectivity is required. This is essential for students or users with limited or unstable network access.

CONTEXT LEVEL AND DATA FLOW DIAGRAM

Context Level Diagram (Level 0):



Data Flow Diagram -(Level 1)



SYSTEM SPECIFICATION

1. Hardware Specifications:

Development Environment:

- **Processor:** Intel Core i3/i5 (or AMD Ryzen 3/5 equivalent)
- **RAM:** 4 GB (minimum); 8 GB (recommended) for smoother browser performance
- **Storage:** 120 GB SSD/HDD (minimum); 256 GB SSD or higher (recommended)
- **Graphics:** Integrated GPU (sufficient for web development).
- **Internet:** Not mandatory for core functionality (offline-capable), but recommended for portfolio import testing.
- **Display:** Minimum 1366×768 resolution; Full HD (1920×1080) recommended for better UI design and preview.

2. Software Specifications:

Development Tools:

- **Operating System:** Windows 10/11.
- **IDE/Editor:** Visual Studio Code (preferred) or any text editor.
- **Version Control:** Git with GitHub/GitLab for code management.
- **Libraries Dependencies:** CDN-based JavaScript libraries.

Frameworks and Libraries:

- **Frontend:** HTML5, CSS3, JavaScript (ES6+)
- **React:** React.js (UMD version), React DOM UMD
- **File Handling & Utility Libraries:** Chart.js, CSV Parsing

Browsers:

- Google Chrome, Firefox, Safari or Edge (latest versions for development and testing).

Additional Tools:

- **Excel / Google Sheets:** For preparing mutual fund CSV files for import.
- **Browser Dev Tools:** Debugging UI and Local Storage data.

TOOLS USED

The successful development of FinTrack – Mini Finance App was achieved by using a combination of front-end technologies, browser-based tools, visualization libraries, and version control systems. Below is a detailed explanation of the tools used and their roles in this project:

1. **HTML5:** Used to create the structural layout of the web application. It helped design various sections such as Transactions, Investments, Liabilities, and SIP Projections using forms, tables, input controls, and organized page navigation.
2. **CSS3:** Used to style the entire interface and ensure a clean, responsive, user-friendly layout. Flexbox and Grid layout helped in creating a structured, dashboard-style layout. Custom styling was applied for dark mode theme support.
3. **JavaScript (ES6+):** Served as the core scripting language for implementing application logic such as calculations, form handling, data persistence, filtering, dynamic rendering, graph analysis, liability coverage checking, SIP projection, and mutual fund calculations.
4. **Chart.js:** A JavaScript charting library used to visualize financial data such as category-wise expenses, investment allocation, and income-expense trends. It enhanced financial insights with interactive pie and bar charts.
5. **React.js:** Lightweight usage to demonstrate component-based UI implementation for theme toggling and dynamic chart rendering. It helped showcase modern UI practices without setting up a full React application.
6. **Git & GitHub:** Git was used for version control to track project development progress, while GitHub repositories were used for storing source code, sharing project files, and collaborating.
7. **VS Code:** The primary code editor used for writing, debugging, and managing HTML, CSS, and JavaScript files. Extensions like Live Server, Prettier, and JavaScript IntelliSense provided efficient development support.

SAMPLE SCREENSHOTS

NOTE: The project has 2 different display modes: Light mode and Dark mode

Transactions tab:

FinTrackLIGHT

TRANSACTIONSINVESTMENTSLIABILITIES

ADD TRANSACTION

Amount

Expense

Category

dd-mm-yyyy

Note (optional)

ADD

RESET

Balance: ₹2330.00

Income: ₹6600.00

Expense: ₹4270.00

Search note or category...

Date ↓

All time ↓

EXPORT SUMMARY

DATE	CATEGORY	NOTE	TYPE	AMOUNT	
2025-11-24	Liability	Payment for liability: Bus pass fee	Expense	300.00	DELETE
2025-11-24	Gift	-	Income	1100.00	DELETE

Sorting (by Date or Amount):

FinTrackLIGHT

Search note or category...

Date ↓

Month ↓

November, 2025

EXPORT SUMMARY

DATE	CATEGORY	NOTE	TYPE	AMOUNT	
2025-11-24	Liability	Payment for liability: Bus pass fee	Expense	300.00	DELETE
2025-11-24	Gift	-	Income	1100.00	DELETE
2025-11-22	Salary	-	Income	3000.00	DELETE
2025-11-22	Shopping	-	Expense	800.00	DELETE
2025-11-22	Transport	-	Expense	900.00	DELETE
2025-11-17	Bills	Electricity	Expense	300.00	DELETE
2025-11-16	Education	-	Income	1000.00	DELETE
2025-11-16	Party	-	Expense	200.00	DELETE
2025-11-16	Freelancing	Instagram	Income	500.00	DELETE

Timeline selection (Month, Year, All time):

Search note or category...

Date ↓

Month ↓

November, 2025

EXPORT SUMMARY

DATE	CATEGORY	NOTE	TYPE	AMOUNT	
2025-11-24	Liability	Payment for liability: Bus pass fee	Expense	300.00	DELETE
2025-11-24	Gift	-	Income	1100.00	DELETE

Month Selection Panel:

Search note or category...

Date ▾

Month ▾

November, 2025

EXPORT SUMMARY

DATE	CATEGORY	NOTE	TYPE
2025-11-24	Liability	Payment for liability: Bus pass fee	Expense
2025-11-24	Gift	-	Income
2025-11-22	Salary	-	Income

2025

Jan

Feb

Mar

Apr

May

Jun

Jul

Aug

Sep

Oct

Nov

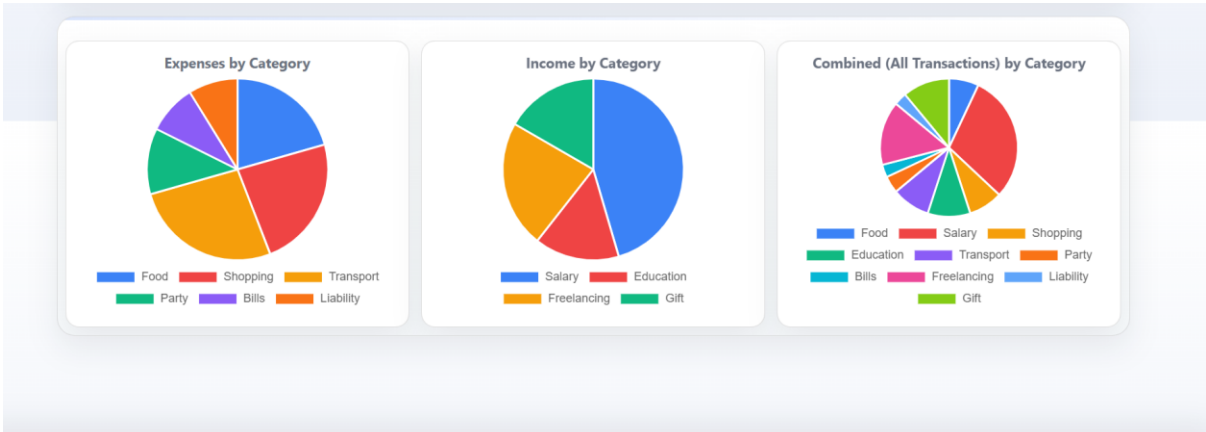
Dec

Clear

This month

Summary can be exported as CSV

React-based Pie charts:



Made by Akshat Goyal
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Dark Mode of Transactions tab:

FinTrack

DARK

TRANSACTIONS

INVESTMENTS

LIABILITIES

ADD TRANSACTION

Amount

Expense ▾

Category

dd-mm-yyyy

Note (optional)

ADD

RESET

Balance: ₹2330.00

Income: ₹6600.00

Expense: ₹4270.00

Search note or category...

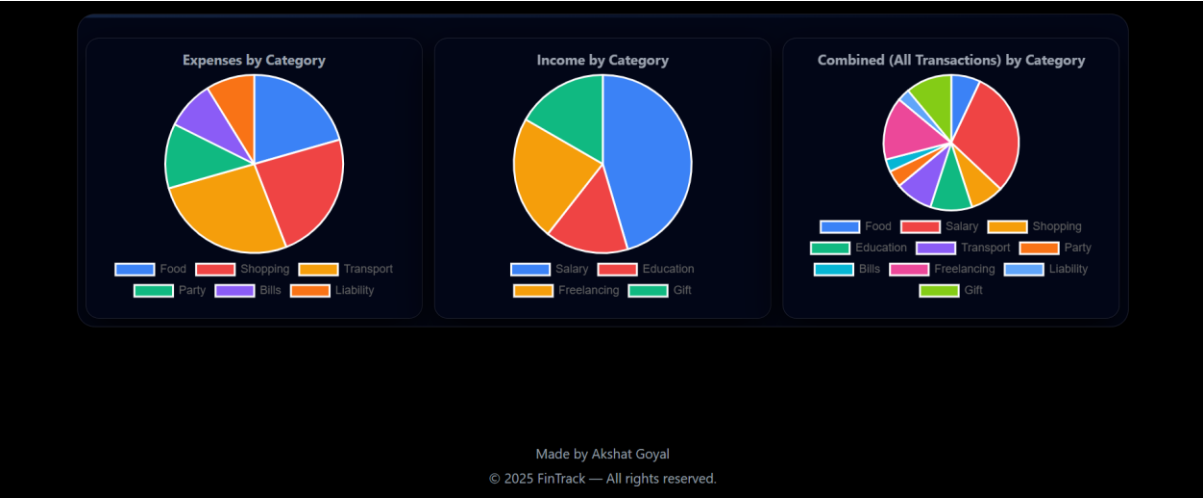
Date ▾

Month ▾

November, 2025

EXPORT SUMMARY

DATE	CATEGORY	NOTE	TYPE	AMOUNT	
2025-11-24	Liability	Payment for liability: Bus pass fee	Expense	300.00	DELETE
2025-11-24	Gift	-	Income	1100.00	DELETE



Investments tab:

Import Portfolio, Holdings Displayed:

FinTrack DARK

TRANSACTIONS **INVESTMENTS** LIABILITIES

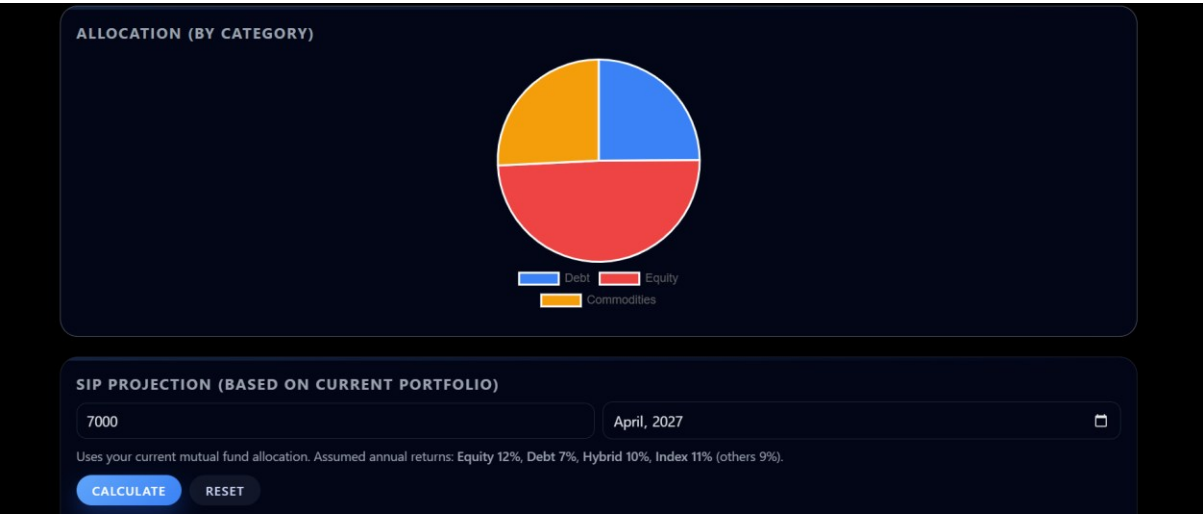
INVESTMENTS (BASIC)

Import Portfolio (CSV)

HOLDINGS

SCHEME	CATEGORY	UNITS	AVG COST	CURR NAV	VALUE
Edelweiss Liquid Direct Growth	Debt	0.5750	3487.57	3487.5652	2005.35
ICICI Prudential Infrastructure Direct Growth	Equity	4.5560	214.16	214.1594	975.71
SBI Gold Direct Plan Growth	Commodities	28.0570	37.68	37.6833	1057.28
ICICI Prudential Nifty 50 Index Direct Plan Growth	Equity	3.7020	273.85	273.8547	1013.81
SBI Silver ETF FoF Direct Growth	Commodities	62.9950	16.12	16.1176	1015.33
Bandhan Small Cap Fund Direct Growth	Equity	19.0150	51.46	51.4620	978.55
SBI Flexicap Fund Direct Growth	Equity	7.9780	125.54	125.5427	1001.58

Pie-Chart:



SIP Projection:

SIP PROJECTION (BASED ON CURRENT PORTFOLIO)

7000

April, 2027

Uses your current mutual fund allocation. Assumed annual returns: Equity 12%, Debt 7%, Hybrid 10%, Index 11% (others 9%).

CALCULATE

RESET

Projection period: ~1.50 years (18 months)

Current portfolio value: ₹8047.61

Monthly SIP amount: ₹7000.00

Total SIP invested (18 installments): ₹126000.00

Projected value (no new SIPs): ₹9283.38 (gain ₹1235.77)

Projected value (with monthly SIPs): ₹144181.17 (gain ₹10133.56)

Fund-wise projection:

Edelweiss Liquid Direct Growth: now ₹2005.35 → ₹35172.41 (existing units: ₹2219.55, SIP part: ₹32952.86, weight 24.9%)

SBI Gold Direct Plan Growth: now ₹1057.28 → ₹18810.99 (existing units: ₹1203.18, SIP part: ₹17607.82, weight 13.1%)

ICICI Prudential Nifty 50 Index Direct Plan Growth: now ₹1013.81 → ₹18422.10 (existing units: ₹1201.67, SIP part: ₹17220.44, weight 12.6%)

SBI Flexicap Fund Direct Growth: now ₹1001.58 → ₹18199.87 (existing units: ₹1187.17, SIP part: ₹17012.70, weight 12.4%)

SBI Silver ETF FoF Direct Growth: now ₹1015.33 → ₹18064.62 (existing units: ₹1155.44, SIP part: ₹16909.19, weight 12.6%)

Bandhan Small Cap Fund Direct Growth: now ₹978.55 → ₹17781.39 (existing units: ₹1159.87, SIP part: ₹16621.51, weight 12.2%)

ICICI Prudential Infrastructure Direct Growth: now ₹975.71 → ₹17729.78 (existing units: ₹1156.51, SIP part: ₹16573.27, weight 12.1%)

Note: This is a simple projection using assumed annual returns by category, your current mutual fund holdings, and a fixed monthly SIP amount. Actual market returns will differ.

Liabilities Tab:

Liability Entering, Planned Incomes:

FinTrack

LIGHT

TRANSACTIONS

INVESTMENTS

LIABILITIES

ADD LIABILITY

Liability name

Amount (₹)

dd-mm-yyyy

Priority: Low

☐ Allow using mutual funds if short

Note (optional)

ADD LIABILITY

RESET

PLANNED INCOMES (FOR LIABILITIES)

Amount (₹)

dd-mm-yyyy

Note (optional)

ADD PLANNED INCOME

RESET

DATE	AMOUNT	NOTE	
2025-12-15	1100.00	Gifts	DELETE

LIABILITIES OVERVIEW

Coverage is calculated using your current balance, planned incomes before each due date, and optionally mutual fund value if enabled.

NAME	AMOUNT	DUE DATE	PRIORITY	MF ALLOWED	COVERAGE	STATUS
------	--------	----------	----------	------------	----------	--------

Liabilities Record:

LIABILITIES OVERVIEW						
Coverage is calculated using your current balance, planned incomes before each due date, and optionally mutual fund value if enabled.						
NAME	AMOUNT	DUE DATE	PRIORITY	MF ALLOWED	COVERAGE	STATUS
Shoes payment	7000.00	2025-11-30	Low	No	Short by ₹4670.00	open
Clothes	8000.00	2025-12-12	Low	Yes	Paid	paid
Parking fee	100.00	2025-12-13	Low	Yes	Covered	open
Short trip	1000.00	2025-12-25	Low	No	Covered	open

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OUTPUT REPORTS

This section outlines the key reports generated during the development and implementation of the FinTrack - Mini Finance App.

- a) Financial Summary Report (CSV Export)
 - Includes date, category, type, note, and amount
 - Shows monthly/yearly income, expenses, and balance.
- b) Investment Allocation Analysis
 - Auto-generated portfolio distribution graph
- c) Liability Coverage Analysis
 - Displays covered / shortfall based on due date, planned incomes, and MF value.
- d) Projected Portfolio Value Report
 - Year-month selection
 - Provides future estimated MF value and growth without needing SIP contributions.

CONCLUSION

The development of FinTrack – Mini Finance App has been a comprehensive journey that combines financial logic, data visualization, UI/UX design, and browser-based data persistence. The primary objective of this project was to build a user-friendly, efficient, and visually interactive application that enables individuals to track their expenses, income, investments, liabilities, and SIP projections in a single integrated platform. The system was carefully designed to ensure clarity, automation, and insightful financial decision-making, without requiring complex backend infrastructure.

One of the key accomplishments of this project is the dynamic and responsive interface developed using HTML, CSS, and JavaScript, ensuring an intuitive and smooth user experience. Features such as automatic balance calculation, real-time investment valuation, liability coverage analysis, SIP projections, and CSV-based portfolio import demonstrate the practical usability of the application in real-life financial scenarios. The use of Local Storage successfully enables secure and persistent data handling directly within the browser, eliminating the need for external databases.

The project also emphasizes scalability, reusability, and modular development, ensuring that the system can be expanded in the future to include advanced features such as AI-based financial advisors, tax computation, goal-based planning, loan tracking, and cloud database integration. Visualization using Chart.js and selective use of React.js improves clarity and engagement, making financial analysis more understandable for general users.

Overall, FinTrack demonstrates how modern web technologies can be combined to build a powerful, interactive, and practical finance management system that helps users gain better control over their money. The project successfully meets its intended objectives and lays a strong foundation for further development, reinforcing the importance of digital financial literacy and smart money management in the modern age. This achievement reflects not only technical proficiency, but also the ability to design real-world solutions that promote financial awareness and personal economic empowerment.